

Shoulder Complex

→ Shoulder Complex considered as 4 subpoints

1) Sternoclavicular Joint

2) Acromioclavicular Joint

3) Scapulothoracic Joint

4) Glenohumeral Joint

* Shoulder complex
Composed of the
clavicle, scapula
and humerus.

① Sternoclavicular Joint :-

→ It is a plane synovial joint

3° - rotators

3° - translatory.

→ Joint has a synovial cavity, a joint disc & 3 ligaments

→ Clavicle is linked with lateral end of clavicle at the
scapula
acromioclavicular joint.

Sternoclavicular Articulating Surfaces :-

→ It consists of 2 shallow, saddle-shaped surfaces.

Medial → notch formed by the manubrium of the
sternum & 1st costal cartilage.

Lateral → Medial end of the clavicle.

Sternoclavicular Disc :-

→ It has fibrocartilage disc or meniscus.

→ Upper portion of Sternoclavicular disc is attached to the
posterio-superior clavicle.

→ Lower portion of Sternoclavicular disc is attached to the
manubrium and first costal cartilage.

→ Shoulder motion - disc act like hinge & pivot for the medial end of the clavicle.

+ During Elevation and depression - medial articular surface rolls & slides upper attachment of disc - pivot point.

+ During Protraction and Retraction - Sternoclavicular disc and medial articular surface roll and slide together on the manubrial facet lower attachment of disc - Pivot point.

→ Part of Manubrium - Elevation & depression

→ part of clavicle - Retraction & protraction

→ Disc provides - Stability by ring joint congruence

Sternoclavicular joint Capsule & Ligament :-

→ It is surrounded by a fairly strong fibrous capsule but depends on three Ligaments

① Anterior & Posterior Sternoclavicular Ligament

② Bilaminar Costoclavicular Ligament

③ Interclavicular Ligament.

① Sternoclavicular Ligament :-

→ It provides anterior & posterior sternoclavicular ligament.

+ Anterior and posterior sternoclavicular ligament check anterior and posterior translatory movement of the medial end of the clavicle.

② Costoclavicular Ligament :-

- It is very strong ligament b/w clavicle & 1st rib.
- They have 2 Laminae.

1) Anterior laminae - It has fibers directed laterally from 1st rib to the clavicle.

2) Posterior laminae - Directed medially from rib to the clavicle.

③ Interclavicular Ligament :-

- It resists excessive depression of the distal clavicle and superior glide of the medial end of the clavicle.

Sternoclavicular Motions :-

- 3° - rotary of degree of freedom.

Elevation / Depression

Protraction / Retraction

Anterior / posterior rotation

Elevation and Depression of the clavicle :-

- Occurs in AP axis, b/w convex clavicular surface Concave surface by manubrium & the first costal cartilage.

→ Elevation - Lateral clavicle rotates upward.

Depression - Lateral clavicle rotates downward.

- ROM of elevation - 48° (up to)

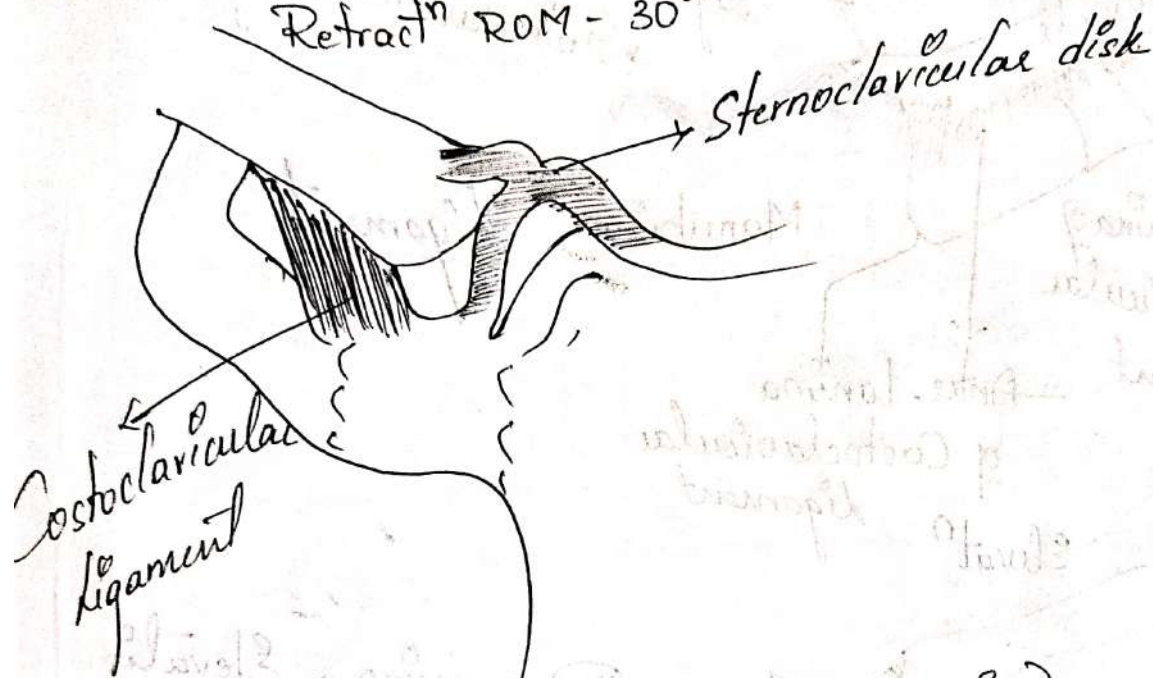
ROM of Depression - less than 15°

Protraction and Retraction of the clavicle :-

- Occur in vertical axis (Superoinferior).
- Protraction - Lateral clavicle moves anteriorly
- Retraction - Lateral clavicle moves posteriorly

Protractⁿ ROM - 15° to 20°

Retractⁿ ROM - 30°



Anterior and Posterior Rotation - (long axis)

- Rotation of the clavicle occurs as a spin b/w the saddle-shaped surfaces of the medial clavicle & manubriocostal facet.

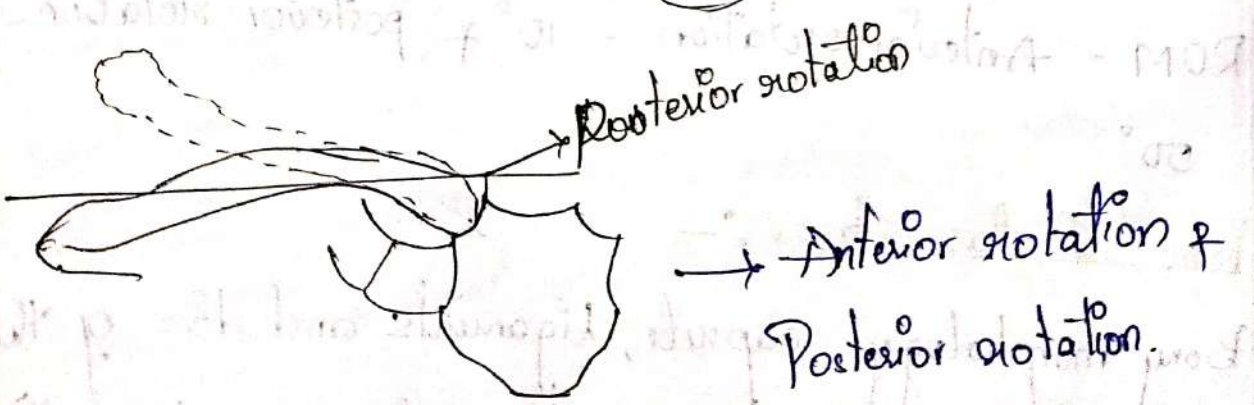
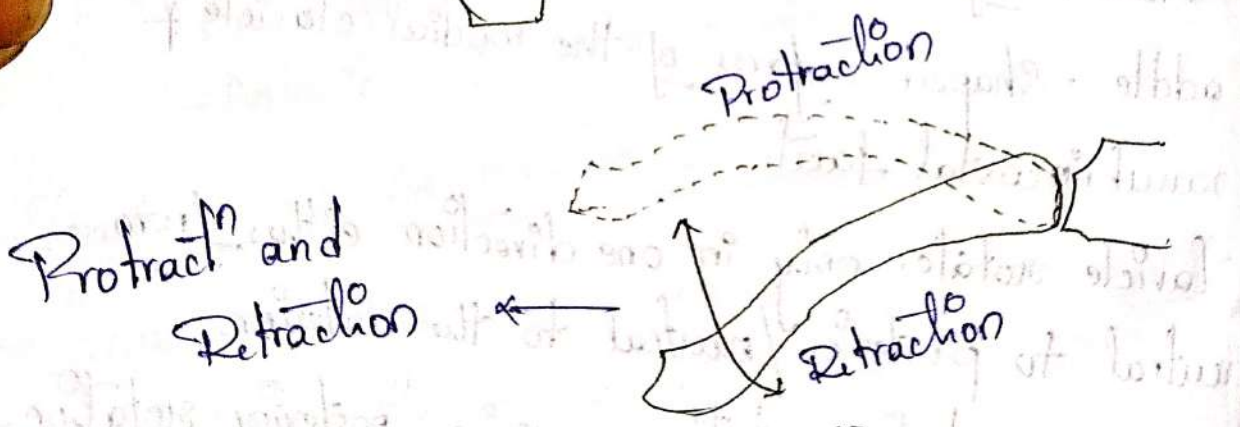
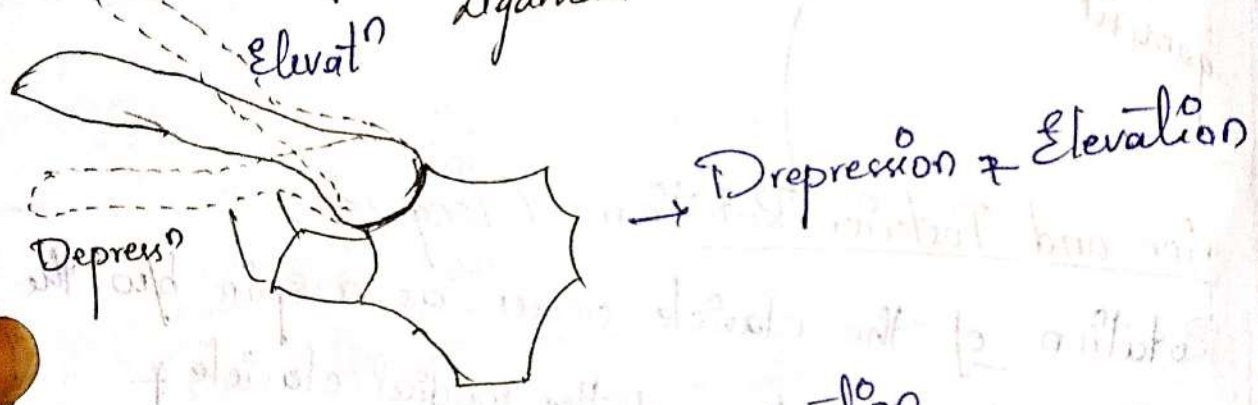
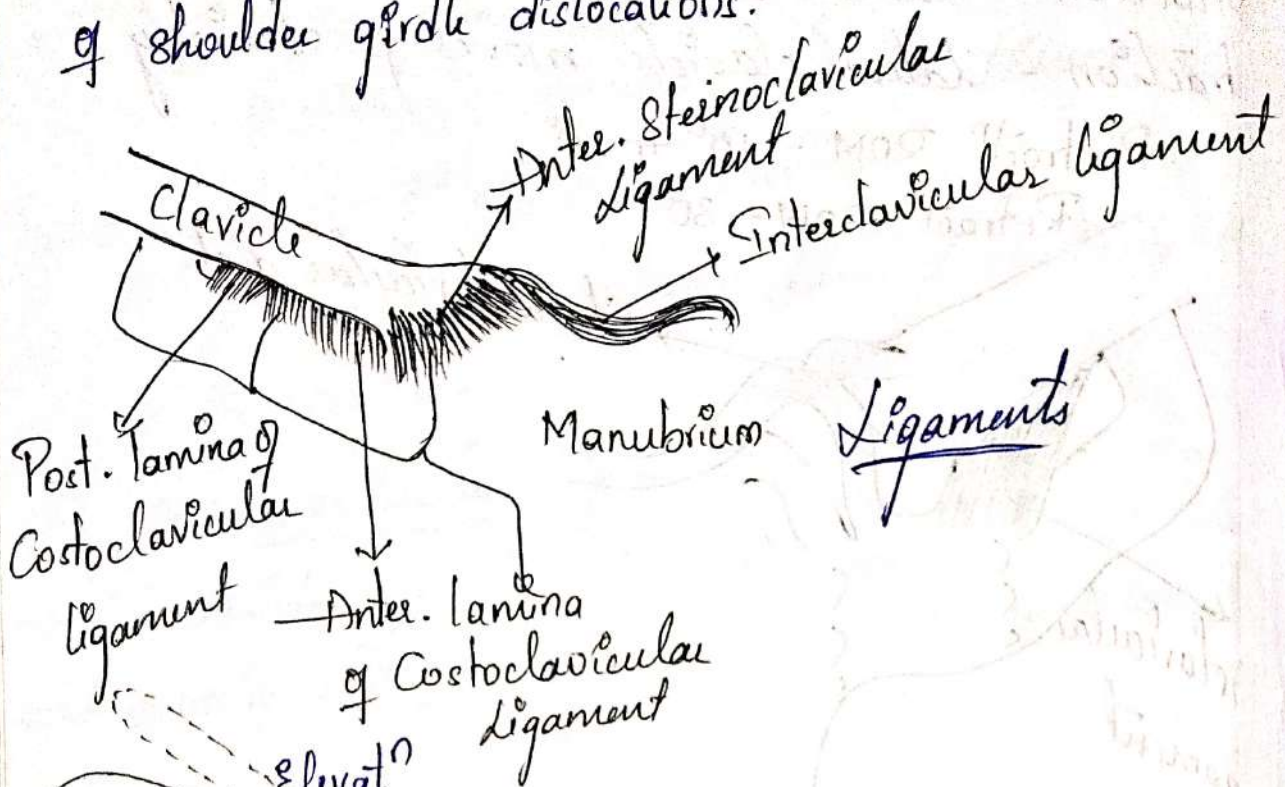
- Clavicle rotates only in one direction either from neutral to posterior / neutral to the anterior.

- ROM - Anterior rotation - 10° & posterior rotation - 50°

Sternoclavicular Stress :-

- Bony morphology, capsule, ligaments and disc of the sternoclavicular joint combine to create an articulation.

→ Dislocation of the sternoclavicular joint represent only 1% of joint dislocations in the body and only 3% of shoulder girdle dislocations.



Shoulder Complex

Scapulothoracic Joint :-

- They are interconnected or interlinked joint.
- 2nd to 7th rib.
- Internally rotated - 35° / 45° / frontal or coronal plane.
- Anteriorly tilted - 10° to 15° / Vertical axis
- Upward rotated - 5° to 10° / Vertical axis

Motions of the Scapula :-

- Anterior and posterior tilting } Secondary Scapular movement
- Internal & External rotation }
- Upward & Downward rotation → Primary Scapular movement
- 2 translatory motion.
 - Elevation & depression
 - Protraction & Retraction

Upward and Downward Rotation -

- It is seen during the elevation of the arm of more significant role in increasing the arm's range of elevation overhead.
- Nearly 50° to 60° of upward rotation of the scapula on the thorax is typically available.
- It is a closed kinematic chain relationship of sternoclavicular and acromioclavicular motions.

Elevation and Depression -

- It is translatory movement
- No change in the degrees, only the scapula moves upward & downward over thoracic cage.

Protraction and Retraction -

- In protraction - Scapula moves away from vertebral column. (Abduction)
- In Retraction - Scapula moves towards from vertebral column (Adduction)
- It is also known as Abduction and Adduction respectively.

Internal and External Rotation -

- Not easily observable.

→ These movements of the scapula on thorax normally accompany protraction & retraction of clavicle at the Sternoclavicular joint.

- Excessive internal rotatⁿ of scapula results winging of scapula.

Anterior and posterior tilting -

- It occurs at acromioclavicular joint.

- Not observed clearly

- Coupled with elevatⁿ & depression of clavicle at the sc^{apula} Sternoclavicular jt.

Scapulothoracic Stability -

- It have dynamic stability and also
- Stabilization is provided by muscles.

Acromioclavicular joint

- Acromioclavicular jt attaches the scapula to the clavicle.
- It is a plane synovial jt.
- 3° rotation
 3° translation. } of freedom.

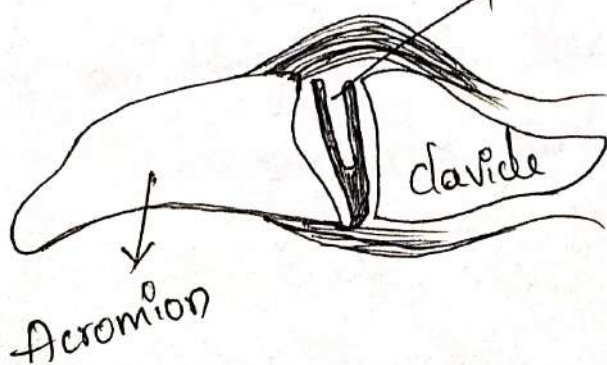
→ It contains jt capsule, 2 major ligaments, joint disc

Acromioclavicular Articulating Surface :-

- It articulates b/w the lateral end of clavicle & a small facet on the acromion of scapula.
- Vertical inclination of the articulating surfaces varies from individual to individual.
- Angle of inclination b/w 16° to 36° from vertical.

Acromioclavicular jt Disc :-

- It varies in size b/w individuals, within an individual as they age, & b/w shoulders of the same individual.
- Through 2 yrs age, it is actually a fibrocartilaginous union.



Acromioclavicular Capsule and Ligaments :-

- Capsule of the ~~ac~~ acromioclavicular jt is weak.
- It cannot maintain integrity of the joint without the reinforcement of the superior and inferior acromioclavicular ligament.

1) Acromioclavicular Ligament.

- Inferior Acromioclavicular Ligament
- Superior Acromioclavicular Ligament.

2) Coracoclavicular Ligament.

- Superior Acromioclavicular Ligament is a main ligament.

- The fiber reinforced by trapezius and deltoid muscle.
- Making superior jt more stronger than inferior jt.

- Coracoclavicular Ligament -
divided into 2 portion -

- Medial portion - Conoid Ligament
- Lateral portion - Trapezoid Ligament.

- 

Coracoid process.

→ Coroid Ligament -

medial & slightly posterior to the trapezoid
Ligament is more triangular & vertically oriented.

→ Trapezoid Ligament -

is quadrilateral in shape & is nearly horizontal
in orientation.

→ Medial and lateral portions are separated by adipose
tissue and a large bursa.

Acromioclavicular Motions :-

Internal rotation /

External rotation

Anterior / posterior tilting

Upward / downward rotation

→ Internal / External rotation - vertical axis

→ Anterior / Posterior rotation - Oblique Coronal axis

→ Upward / downward rotation - Oblique Anteroposterior axis.

Internal and External rotation :-

→ Internal & External rotation of Scapula in relation
to the clavicle. Occurs along the vertical axis
through the acromioclavicular joint.

→ Internal & External rotation can be visualized by
holding the glenoid fossa of the scapula

anteromedially & posteriolaterally, respectively.

Anterior & Posterior tilting :-

- Occurs through coronal axis of the joint.
- Elevation of Scapula on thorax can result anterior tilt. Ex :- Struggling of shoulder.
- During flexion and abduction the scapula posterior tilt on thorax as it is upwardly rotating.
- ROM - Nearly 60°

Upward and downward rotation :-

- Occur in antero-posterior axis.
- Upward rotation tilts the glenoid fossa.
- Downward rotation tilts to the opposite motion.

Glenohumeral joint

→ It is a ball and socket synovial joint.

→ 6° of freedom

(3 rotatory and 3 translatory movements)

Articulating Surfaces :-

→ Proximal : Glenoid fossa of the scapula.

→ Distal : Humeral head.

→ Glenoid fossa is tilted slightly upward or downwards during rest.

→ Glenoid fossa is anteverted and retroverted

Anteverted - 10°

Retroverted - 6 to 7°

→ The concavity of the fossa is increased by articular cartilage that is thinner in the middle & thicker on the periphery. Thus improves the congruence with humeral head.

→ Head of humerus is larger than that of glenoid articular surface, forming $\frac{1}{3}$ rd of the sphere.

→ The head faces medially, superiorly and posteriorly with regard to the shaft of the humerus.

→ Angle of torsion is formed by an axis through the humeral head & neck in relation to an axis through the humeral condyles.

→ Angle of inclination is formed by an axis through the humeral head and ~~the~~ neck in relation to a longitudinal axis through the shaft of the humerus and is normally between 130° to 150° in the frontal plane.

→ Posterior torsion - posterior orientation of the humeral head with regard to the humeral condyles.

→ Posterior torsion is also called Retro torsion or Retroversion.

Glenoid Labrum :-

→ The total available articular surface of the glenoid fossa is enhanced by the glenoid labrum.

→ It is fibrocartilaginous tissue within the glenoid cavity of the shoulder joint.

→ It is to provide stability and shock absorption within the joint.

→ It is continuous with

Superiorly - tendon of the long head of biceps brachii

Anteriorly - Anterior band of the inferior glenohumeral ligaments

Middle - Glenohumeral ligament.

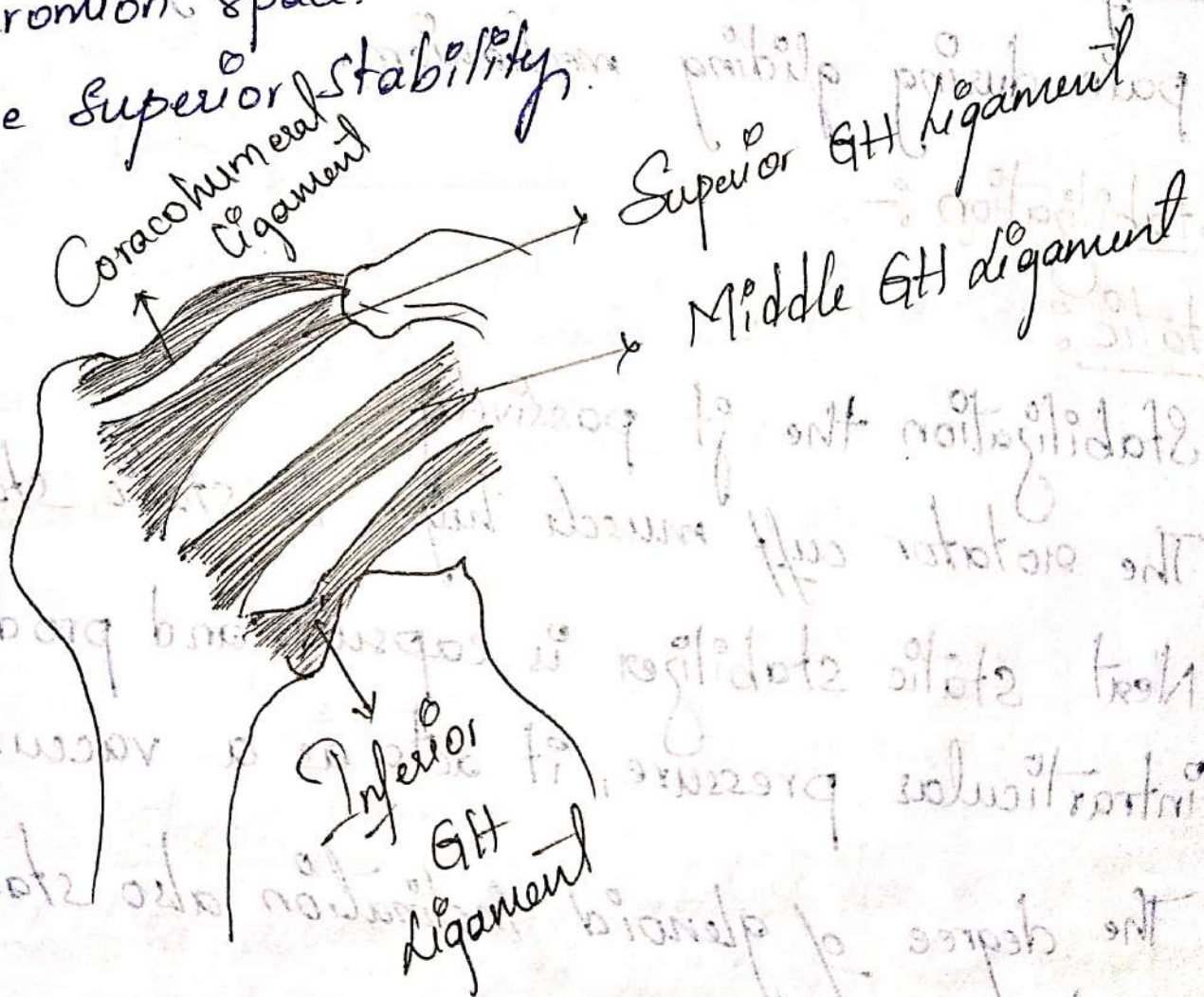
Glenohumeral Capsule and Ligaments :-

- Glenohumeral jt is surrounded by a large, loose capsule that is taut superiorly and slack anteriorly and inferiorly with the arm dependent at the side.
- Capsule is tighten when the humerus is abducted and laterally rotated.
- Provides little stability without the reinforcement of ligaments & muscles.
- The capsule is reinforced by the superior, middle & inferior glenohumeral ligaments.
- Rotator cuff tendon blending with capsule to give stability.
- Superior glenohumeral ligament passes from the superior glenoid labrum to upper neck of the humerus.
- Middle glenohumeral ligament runs obliquely from superior anterior labrum to the anterior proximal surface of the humerus, provides stability.
- Inferior glenohumeral ligament consists of 3 bands
 - * provides anterior stability during lateral rotation.
 - * Inferior stability & medial stability during medial rotation.
- Coracohumeral ligament originates from the base of the coracoid process & has 2 bands.

- * Provides inferior stability.
- * Provides superior stability when rotator cuff damaged.

Coracoacromial Arch :-

- Formed by coracoid process, coracoacromial ligament, the acromion.
- The space b/w head of humerus & arch is called Subacromion space.
- provide superior stability.



Bursae :-

→ Subacromial bursa

→ Subdeltoid bursa

* Function is reduces the friction b/w the articulating surfaces, provides gliding.

* Subacromial bursa permits gliding & reduces the friction.

* Inflammation to bursa lead to swelling or causing pain during gliding mechanism.

Stabilization :-

Static :-

→ Stabilization the jt passively.

→ The rotator cuff muscle helps in static stabilization.

→ Next static stabilizer is capsule and produces intrarticular pressure, it acts as a vacuum.

→ The degree of glenoid inclination also stabilizes the jt.

→ In heavy loaded arm the supraspinatus tendons provides assistance.

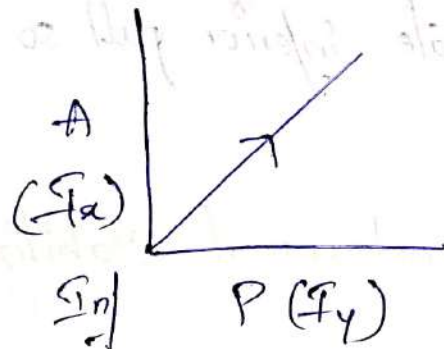
Dynamic :-

1) Deltoid & glenohumeral stabilization :-

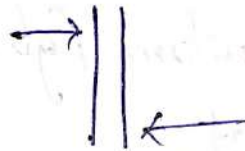
→ Deltoid muscle is major prime mover for abduction.

→ Anterior fibers of deltoid act as prime mover during flexion.

- The action line of 3 fibres segments comes to middle fibres.
- When muscle force vectors, are resolved into parallel (F_x) effects and perpendicular (F_y) component. Superior fibres are far larger than inferior fibres.
- Hence it causes the humeral head to translate superiorly:



- The parallel force component create a shear force rather than compressive force.



Shear force

- Hence deltoid alone can't abduct the arm, another force is required synergistically to produce desired rotations like rotator cuff.

2) Rotator cuff & glenohumeral Stabilization:

- SITS tendons blend & reinforce the glenohumeral Capsule.
- The forceful vectors of all the SITS components it can be seen than perpendicular & gives compression force.

→ Generate compressive force

Infraspinators
Teres minor
Subscapularis } Major jt compressors.

→ Infraspinatus, teres minor are lateral rotators with abduction.

→ This cuff muscles create inferior pull so the head is stabilized.

3) Supraspinatus & glenohumeral stabilization:

→ Superior stability.

→ Superior translatory component rather than inferior translatory component like other rotator cuff.

→ Supraspinatus has large momentum (glenohumeral abduction) and stabilizing the jt.

4) Long head of biceps & glenohumeral stabilization:

→ Anterior stabilization.

→ It capsule is slack at 2 parts anteriorly & inferiorly.

→ Contributes the stabilization by altering the head in the fossa by reducing vertical i.e., Superior, inferior & anterior translations.

Scapulohumeral Rhythm:-

- The full range of shoulder motion normally is combinatⁿ b/w the motion in GH and scapulo-thoracic jt.
- Natural rhythm/ratio b/w GH & ST jt is 2:1
- ROM - Shoulder abduction/flexion ~~is~~ 180°
 120° shoulder abductⁿ/flexion (GH jt)
 60° scapular upward rotation (ST jt)
- Without scapula rotation, the humerus would not be able to attain full ROM.

Glenohumeral Motions :-

Osteokinematics -

Flexion / Extension -

- Sagittal plane and frontal axis
- ROM - 120° of flexion
 50° of Extension

Abduction / Adduction -

- Frontal plane and sagittal axis
- ROM - 0° - 180° of Abduction
 180° - 0° of adduction

Medial / Lateral Rotation -

- Transverse plane and vertical axis

ROM - 70° of medial rotation
 90° of lateral rotation.